CS506: Quantum Computing

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Topic: Tight Analysis of Grover's Search Algorithm
04/29/2026

1 Introduction and Problem Setup

Here we perform a precise mathematical tight analysis of Grover's Algorithm to determine the optimal number of queries and the probability of success. We assume a search space of size $N = 2^n$, where t items satisfy the search criteria.

We define the function $f : \{0, 1, \ldots, N-1\} \to \{0, 1\}$ such that:

- $f(x) = 1$ if x is a "good" state (a solution). There are exactly t such states.
- $f(x) = 0$ if x is a "bad" state. There are $N - t$ such states.

2 Defining the Quantum States

To analyze the algorithm geometrically, we group the basis states into two normalized vectors representing the "good" and "bad" subspaces.

2.1 Good and Bad States

The normalized superposition of all good states is defined as:

$$|\psi_{good}\rangle = \frac{1}{\sqrt{t}} \sum_{x:f(x)=1} |x\rangle \tag{1}$$

Similarly, the normalized superposition of all bad states is:

$$|\psi_{bad}\rangle = \frac{1}{\sqrt{N-t}} \sum_{x:f(x)=0} |x\rangle \tag{2}$$

2.2 The Initial State

The algorithm begins by applying Hadamard gates to the zero state, creating an equal superposition of all N possible states:

$$|\psi\rangle = \frac{1}{\sqrt{N}} \sum_{x\in\{0,1\}^n} |x\rangle \tag{3}$$

We can rewrite $|\psi\rangle$ in terms of our good and bad vectors. By calculating the coefficients, we find that:

$$|\psi\rangle = \sqrt{\frac{t}{N}}\, |\psi_{good}\rangle + \sqrt{\frac{N-t}{N}}\, |\psi_{bad}\rangle \tag{4}$$

To simplify this, we define an angle θ such that:

$$\sin\theta = \sqrt{\frac{t}{N}} \quad \text{and} \quad \cos\theta = \sqrt{\frac{N-t}{N}} \tag{5}$$

Thus, the initial state is $|\psi\rangle = \sin\theta\, |\psi_{good}\rangle + \cos\theta\, |\psi_{bad}\rangle$.

3 The Grover Operators

Grover's algorithm iteratively applies two main operators: the Oracle and the Diffusion operator. Together, they perform a rotation in the 2D plane spanned by $|\psi_{good}\rangle$ and $|\psi_{bad}\rangle$.

3.1 The Oracle (U_f)

The oracle marks the good states by flipping their phase. Geometrically, this is a reflection across the $|\psi_{bad}\rangle$ axis:

$$U_f |\psi\rangle = \cos\theta |\psi_{bad}\rangle - \sin\theta |\psi_{good}\rangle \tag{6}$$

3.2 The Diffusion Operator (D)

The diffusion operator, also known as the inversion about the mean, is defined as $D = 2|\psi\rangle\langle\psi| - I$. When we apply D to the state after the oracle, it reflects the vector across the initial state $|\psi\rangle$.

4 State Evolution and Rotation

Let $|\psi_k\rangle$ be the state after k Grover iterations. After one iteration ($k = 1$), the state becomes:

$$|\psi_1\rangle = \sin(3\theta) |\psi_{good}\rangle + \cos(3\theta) |\psi_{bad}\rangle \tag{7}$$

By induction, after k iterations, the state is:

$$|\psi_k\rangle = \sin((2k+1)\theta) |\psi_{good}\rangle + \cos((2k+1)\theta) |\psi_{bad}\rangle \tag{8}$$

Each iteration rotates the state closer to $|\psi_{good}\rangle$ by an angle of 2θ.

5 Optimal Number of Queries

To maximize the probability of finding a good state, we want the coefficient of $|\psi_{good}\rangle$ to be as close to 1 as possible. This occurs when:

$$(2k+1)\theta \approx \frac{\pi}{2} \implies 2k \approx \frac{\pi}{2\theta} - 1 \implies k \approx \frac{\pi}{4\theta} - \frac{1}{2} \tag{9}$$

Since $\sin\theta = \sqrt{t/N}$, for small θ, we have $\theta \approx \sqrt{t/N}$. Therefore, the optimal number of queries is:

$$k \approx \frac{\pi}{4}\sqrt{\frac{N}{t}} \tag{10}$$

6 Probability Analysis

The probability of success is the square of the amplitude of the good state:

$$P_{success} = \sin^2((2k+1)\theta) \tag{11}$$

The probability of failure is $P_{failure} = \cos^2((2k+1)\theta)$. If we choose k such that it is the closest integer to $\frac{\pi}{4\theta} - \frac{1}{2}$, then the distance between the actual angle and $\pi/2$ is at most θ. Therefore:

$$P_{failure} \leq \sin^2\theta = \frac{t}{N} \tag{12}$$

This gives us a very high success probability: $P_{success} \geq 1 - \frac{t}{N}$.

7 Quantum Circuit and Amplitude Updates

The overall circuit architecture begins with initialization followed by a series of blocks, where each block consists of O_f and D.

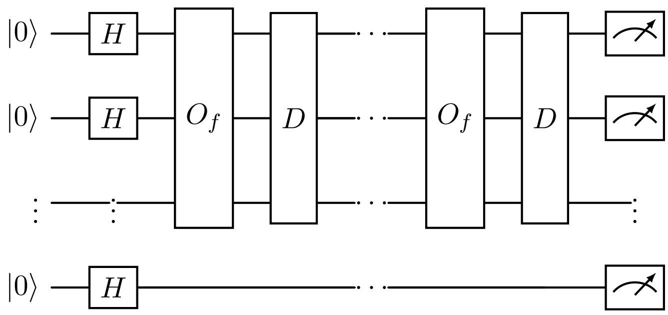

Figure 1: The quantum circuit for Grover's search algorithm.

We track the amplitudes of a good state (α) and a bad state (β). If the mean amplitude is $\mu = \frac{\sum \alpha_x}{N}$, the diffusion operator updates an amplitude α_x to:

$$\alpha_x^{(t+1)} = 2\mu^{(t)} - \alpha_x^{(t)} \tag{13}$$

Specifically, for a good state amplitude α and bad state amplitude β:

$$\alpha^{(t+1)} = 2\mu^{(t)} - (-\alpha^{(t)}) = 2\mu^{(t)} + \alpha^{(t)} \tag{14}$$

$$\beta^{(t+1)} = 2\mu^{(t)} - \beta^{(t)} \tag{15}$$

This shows how the amplitude of the marked state grows relative to the others through each iteration.